

CS & IT ENGINEERING



Algorithms

Sorting Algorithms



Lecture No.- 01

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

TC analysis

Topics to be Covered



Topic

Topic

Topic

Sorting Algos

Intro

Imp terminologies

Bubble Sort



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



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hw

return value $K \rightarrow O()$

loop runs $\rightarrow$ $n \log n$ times

★ $K = K + n/2$

$$K = O\left(\frac{n}{2} \times n \log n\right) = \underline{O(n^2 \log n)}$$



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Sorting

A process of ordering/arranging the given element in a particular sequence/ order as per the given criteria.





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Eg.1. 50, 79, 3005, 452, 297

#Q. Arrange the given element in increasing order of their values?

→	50,	79,	297,	452,	3005
	1	2	3	4	5



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Eg.2. 50, 79, 3005, 452, 297

Arrange in Decreasing order of values.

→ 3005, 452, 297, 79, 50,





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Eg.3. Arrange the given elements in increasing order of their 1st digit.

50,	79,	3005,	452,	279
5 ↓	7 ↓	3 ↓	4 ↓	2 ↓
297,	3005,	452,	50,	79



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Sorting Algorithms:

1. Bubble Sort

2. Selection Sort

3. Insertion Sort

4. Merge Sort

5. Quick Sort

6. Heap Sort

7. Radix Sort



→ Divide & Conquer

→ Heaps



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For every sorting Algorithms:

Example ✓

Code ✓

Internal working ✓

Complexity analysis ✓

- Time
- Space

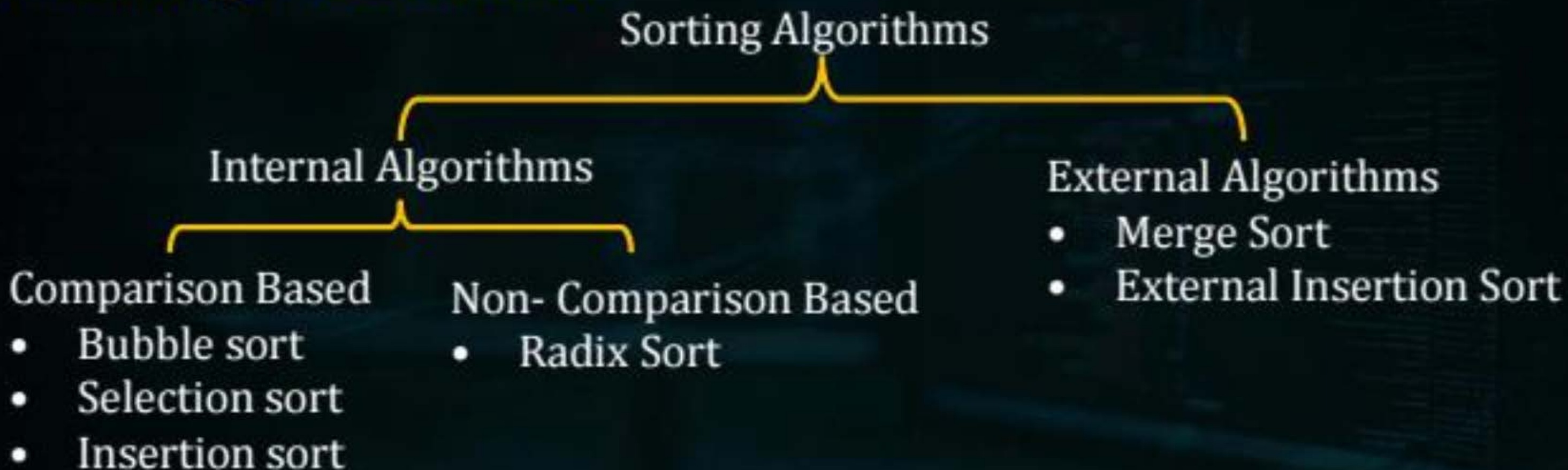
Properties ✓



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Terminologies & Classification:





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Internal Sorting:-

- All the elements should be together present in the main memory.
- Ruffer for smaller input size.

preferred



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External Sorting:-

- All the elements to be sorted, need ^{not} ~~to~~ be present together at once in the main memory.
- ~~Buffer~~ (for larger input size.)
- Requires frequent transfer of data between main memory & secondary memory.





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Comparison-Based Sorting:-

- When an element is compared with the other element of the given data to achieve sorting.



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Non-Comparison-Based Sorting:-

- When sorting is achieved without any comparison among the elements to be sorted.

eg:- Radix Sort



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Other classification of sorting Algorithms:-

1. Internal vs External
2. Comparison vs non-comparison ✓
3. Recursive vs Iterative
4. Inplace vs not-inplace
5. Stable vs unstable.

properties

Note:- If nothing is specified, the default sorting order will be ascending/increasing.



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In-place sorting Algorithm:-

Additional/Auxiliary space Requirement

- At most $O(1)$ {excluding recursion stack}
- Recursion stack of atmost $O(\log n)$

→ inplace Algo

Eg.4. Merge Sort:- Space complexity $\Rightarrow O(n + \log n)$

↓
not inplace

↓
 $B(n)$

↓
Recursion



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Stable sorting Algorithms:-

- If relative position of two equal elements is maintained before & after sorting.



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Formal Definition:-

If $A[i] = A[j]$

&

In given input: $A[i]$ is before $A[j]$

Then, algorithm is stable if $A[i]$ is before $A[j]$ in finally sorted (after sorting) array as well.





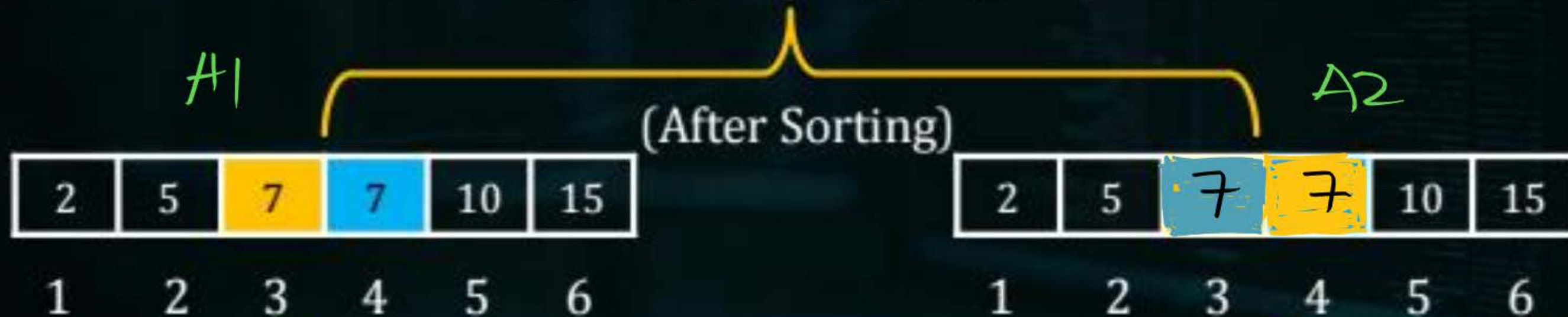
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Eg.5. Stable vs unstable:

Given i/p Array to be sorted

5	10	7	15	7	2
1	2	3	4	5	6

(Before Sorting)



↓
Stable Sorting

↓
unstable
Sorting



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Inversion in an Array

Formal Definition:

- A pair of indices (i, j) is considered to be an inversion of the given Array $A[1...n]$ if $i < j$ and $A[i] > A[j]$

Note - i & j need not be adjacent indices (not mandatory)

In simple terms, if a larger element is present before a smaller element, then their indices will form an inversion pair.



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Eg.6.



$(10, 12) \times$

Total number of inversion pairs in given array A?

<u>(1, 3)</u>	<u>(1,4)</u>	<u>(1,5)</u> →	3
<u>(2, 3)</u>	<u>(2,4)</u>	<u>(2,5)</u> →	3
<u>(3, 4)</u>	<u>(3,5)</u>	→	2
<u>(4, 5)</u>		→	1

} ∴ 9

Total number of inversion = $3+3+2+1 = 9$



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Range of number of inversion in given array:

Given $A [1.....n]$ max min

Min number of inversions

→ A is in increasing order

	1	2	3	4
A:	5	10	15	18



Inv: 0 0 0 0

$N = 4$

Total inversion = 0



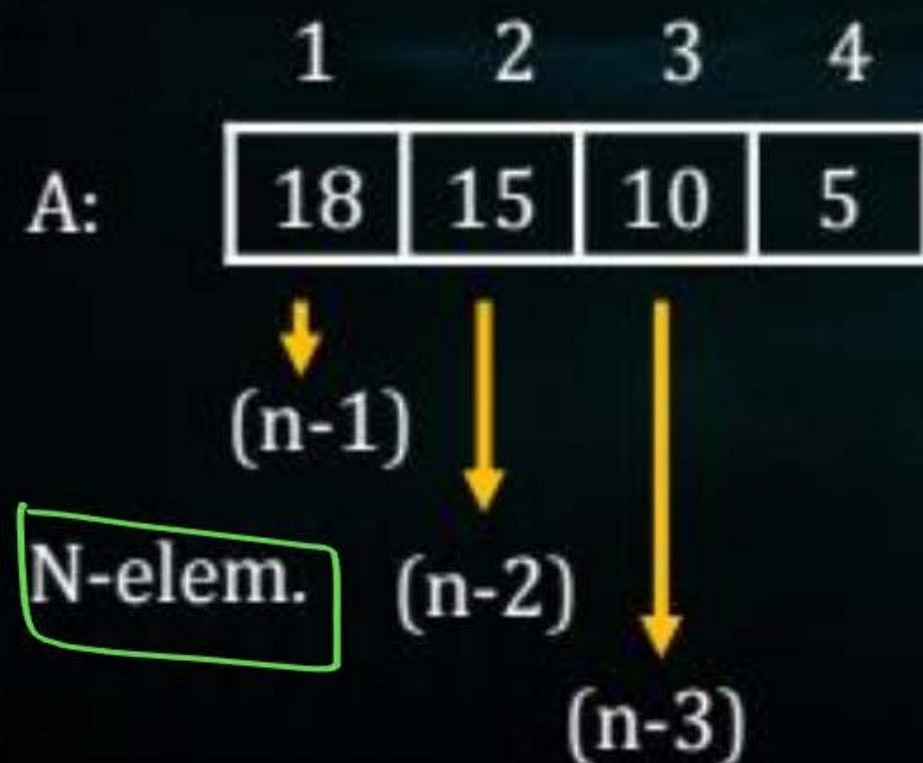
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Range of number of inversion in given array:

Given A [1.....n] ~~max min~~

2. Max number of inversions

→ A is in decreasing order



In general

$$\left(\sum_{i=1}^n i \right) - n$$

$$(n+1) + (n-2) + \dots + 0 = \frac{n(n+1)}{2}$$

$$\begin{aligned} \text{Total} &= (n-1) + (n-2) + \dots + 1 \\ &= \frac{n(n-1)}{2} \end{aligned}$$



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$A \rightarrow n$

$$0 \leq |\text{numberofinversion}| \leq \frac{n(n-1)}{2}$$

$$\left(\sum_{i=1}^n i \right) - n = \frac{n(n+1)}{2} - n = \frac{n^2 + n - 2n}{2} = \frac{n^2 - n}{2} = \frac{n(n-1)}{2}$$



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V. Imp

Time complexity of any comparison based sorting algorithms:-

- depends on two important factors:
 1. Number of comparison:
 2. Number of Swaps (number of inversion).

or interchanges

1. Bubble Sort



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Bubble sort → by default: ascending order

- Basic Algorithm(Algorithm -1)
- Better Algorithm(Algorithm -2)
- Optimised Bubble sort(Algorithm -3)

Logic/ Idea:- In every i^{th} pass/ iteration, place the i^{th} max element at its correct position,



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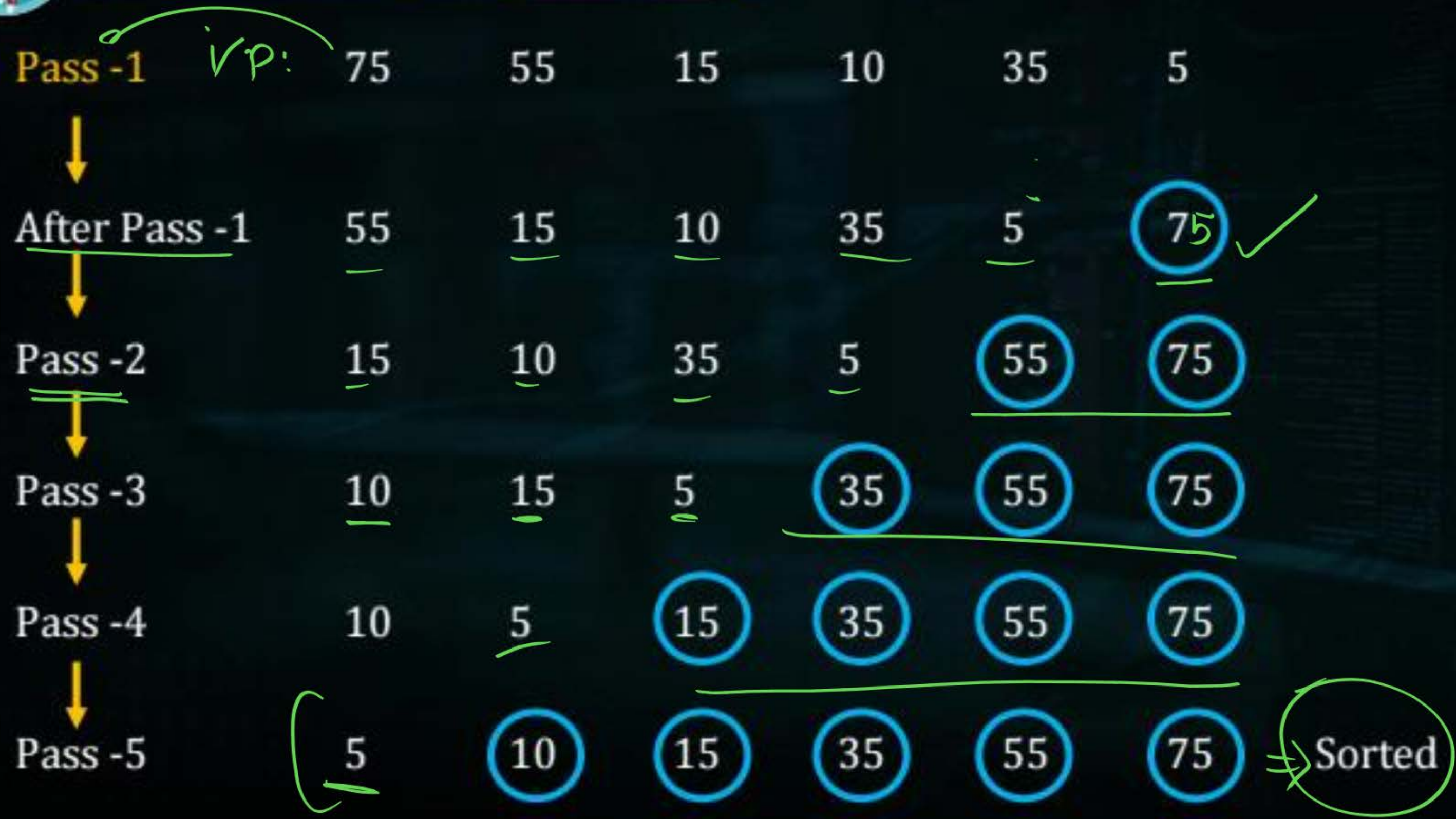
Eg.6. Given

	1	2	3	4	5	6
A:	75	55	15	10	35	5

Given array is input to pass 1



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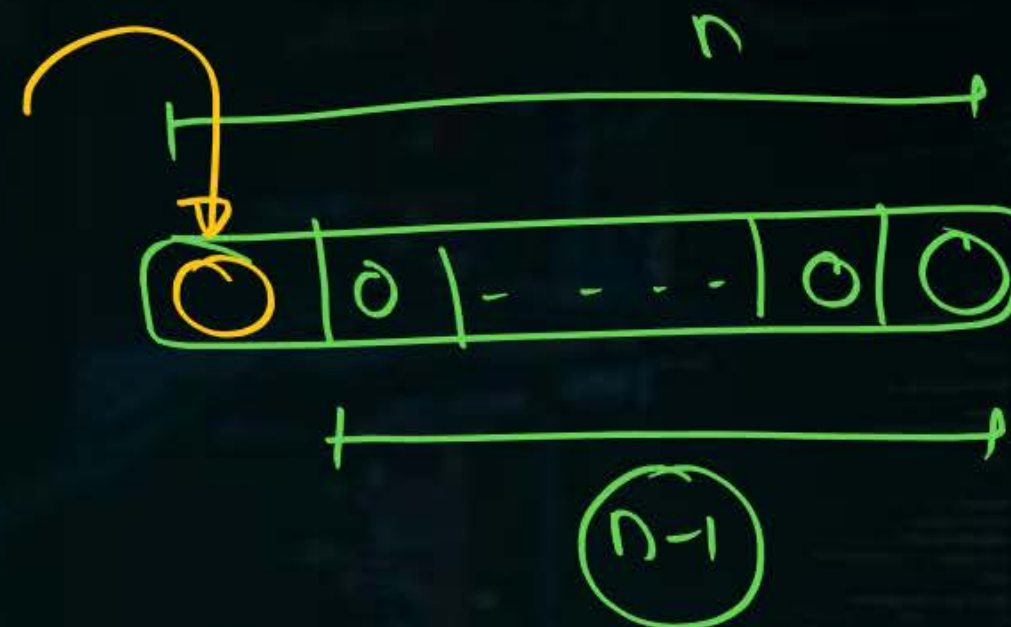


Imp. Observation

After pass 1 $\rightarrow$ 1 elements

After pass 2 $\rightarrow$ 2 elements

After pass $(n-1)^{\text{th}}$ $\rightarrow$ $(n-1)$ elements will be at their correct position.



Note:- The max number of passes required to = $(n-1)$ passes.



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Basic Bubble sort

Algorithm Bubble Sort (A, n)

```
{  
    for (pass = 1 ; Pass <= (n-1) ; Pass ++)  
    {  
        for (j = 1; j <= (n-1) ; j++)  
        {  
            if (A[j] > A[j + 1])  
            {  
                Swap (A[j] , A [j + 1])  
            }  
        }  
    }  
}
```

every
pass $\rightarrow$ (n-1)



Swap(a, b)
{
 t = a
 a = b
 b = t
}



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1. ~~$(n-1)$~~

2. ~~$(n-2)$~~

$$\begin{array}{c} (n-1) \\ + \\ (n-2) \\ + \\ (n-3) \\ + \\ \vdots \end{array}$$

Algorithm time complexity

A is in asc ←

	<u>No. of comparison</u>	<u>No. of Swaps</u>
<u>(inc.) Best case</u>	$(n-1) * (n-1)$ $= (n-1)^2$	0
<u>(dec.) Worst case</u>	$(n-1) * (n-1)$ $= (n-1)^2$	$\frac{n(n-1)}{2}$ $\Rightarrow (n-1) + (n-2) \dots 1$

→ $\Omega(n^2)$

→ $O(n^2)$

$O(n^2)$

A is in desc



2 mins Summary



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Intro

Terminologies

Bubble



THANK - YOU